Review of

Bayesian inference of the climbing grade scale

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1 Summary

The authors implement a Bayesian dynamic Bradley-Terry model for sport climbing ascent data, building on Scarff (2020). The central goal is to estimate the fundamental scale parameter m, which quantifies the proportional increase in difficulty corresponding to one grade increment. Using data from thecrag.com across Australia, New Zealand and Germany, and four grading systems (Ewbank, French, UIAA, V-scale), the authors estimate that one grade increment corresponds to $\approx 2.1\times$ difficulty for sport climbing scales and $\approx 3.17\times$ for bouldering (V-scale). They connect these findings to the Weber-Fechner psychophysical law.

The manuscript addresses a question of genuine interest to the climbing community and to researchers working on subjective rating scales. The core idea of using whole-history rating to infer the scale parameter of a grading system is appealing, and the empirical convergence of estimates across countries and styles is striking and constitutes the paper's most compelling result. That said, in its current form the manuscript has significant gaps in methodological rigor, statistical reporting, and contribution clarity that need to be addressed before it can be considered for publication.

2 Comments

1. The paper repeatedly states it follows Scarff (2020) for both the model and the underlying motivation. The Introduction and Section 3 do not clearly delineate what is genuinely new here. The paper would benefit substantially from an explicit "Contributions" paragraph at the end of the Introduction that distinguishes this work from Scarff (2020).

2. For a paper whose core methodological claim is "full Bayesian MCMC inference," surprisingly little is reported about the inference itself.

- No prior distributions are specified for m, the climber grades $C(t)$, or the Wiener process variance w^2. These should be presented in the main text, accompanied by a sensitivity analysis where appropriate, or at least a justification for them.
- No convergence diagnostics are reported: no R-hat, no effective sample size, no number of chains, no warmup/sampling iterations.

3. The Wiener process variance w^2 in Equation (4) governs how rapidly climber ability is allowed to drift (This should also be stated in the text). Was this estimated, fixed, or assigned a hyperprior?

4. The criterion of $\geq$30 attempts with $\geq$1 explicit failure is reasonable but arbitrary. A robustness check at, say, $\geq$50 and $\geq$100 attempts would help readers gauge how dependent the estimates are on inclusion thresholds. Similarly, the choice of n = 100 climbers is not motivated, why not all eligible climbers? This is especially relevant given the small-n analyses for other countries.

5. The recasting of climbing as a Bradley-Terry (BT) "game" between climber and route is novel enough that it warrants a brief sentence noting that, unlike player-vs-player BT, the route's "ability" R is fixed across the entire data set. This affects identifiability, what anchors the climber-grade scale to the route-grade scale, given route grades are taken as known?

6. The slopes of 0.7–0.79 (Figure 4) "conform remarkably well" with the Bayesian estimates, but as the authors themselves note, this analysis confounds difficulty with the number of attempts the community makes at each grade, and with the supply of routes at each grade. It is not really a validation. I would either drop this analysis or frame it more cautiously, at present it adds rhetorical weight that is not earned by the underlying data.

7. In page 9: "for $n = 100$ climbers of differing abilities", what was the selection rule beyond the $\geq$30 attempts / $\geq$1 failure threshold? Were these the top-100 most-active climbers? Random sample? This matters for selection bias.

8. In Figure 1 the horizontal axis label "Slope" should clarify whether this is $d = e^m$ (per the caption) — the body text variously uses both m and d. You should one for the figure axis and define it clearly.

9. On page 18–19, the authors state that for climbers known to log every attempt, the slope tends to be slightly less than 2 rather than slightly greater than 2, and label this "data not shown." This is arguably the single most important piece of evidence in the paper: it directly addresses the under-reporting bias that the authors acknowledge as the largest threat to their estimates, and would provide an empirical bound on how much the headline $\approx$2.1 figure should be adjusted. This analysis must be included, even briefly, with sample size, selection criteria, and posterior estimates. Without it, the Discussion's claim that the true slope may be slightly less than 2 is unsupported.

10. Page 17: "practising a particle route" should be "practising a particular route"

11. Section 5.1 reuses the Section 4.1 heading ("Estimating the slope of difficulty for the Ewbank climbing grade scale").
12. "Fontainbleau" should be "Fontainebleau"
13. "Ewbanks" appears once on page 9; elsewhere "Ewbank", you should standardize.
14. Equation (5) gives $E[a] = (1 - p_{\text{send}})/p_{\text{send}}$, but the variable a is not introduced. You should describe that a is the number of failed attempts before the first send.
15. You should also pay attention to the use of periods and commas after each equation, as many of them are currently missing.
16. In the abstract "2.1, 2.09 and 2.13" presents inconsistent decimal precision.